

Logging: Working with the Loki Operator

By harnessing the capabilities of logging and the Loki Operator within the s390x architecture, organisations can effectively manage, analyse, and derive actionable insights from log data. This two-part article series explains how this can be done.



Logging plays a pivotal role in modern software development, providing invaluable insights into the performance, health, and behaviour of applications and systems. As containerisation and microservices architectures become increasingly prevalent, efficient log management solutions are essential for maintaining operational excellence. Among these solutions, Prometheus and Loki stand out as robust tools for monitoring and log aggregation, respectively.

Let's first explore the deployment of the logging operator in s390x architecture, along with various use cases and the deployment of a small application for cluster logs. In the next article in this two-part series we will shift our focus to deploying different

applications, examining the dashboard view, and analysing the results.

Understanding logging and Loki Operator

Logging: Logging encompasses the process of capturing, storing, and analysing data generated by applications, servers, and infrastructure components. It serves as a critical tool for troubleshooting, performance monitoring, and security analysis.

Loki Operator: Loki Operator is a Kubernetes operator specifically designed to simplify the deployment and management of Loki. By automating the configuration, scaling, and maintenance of log aggregation infrastructure, Loki Operator streamlines operational workflows

within Kubernetes environments.

Logging architecture overview

Logging within the OpenShift Container Platform involves several key components, each playing a crucial role in collecting, storing, and visualising log data. Figure 1 illustrates the architectural diagram.

Collector: The collector, deployed as a daemon set, resides on each OpenShift Container Platform node. It is responsible for gathering log data from various sources, transforming it as necessary, and forwarding it to specified outputs.

There are two options for the collector: the Vector collector or the legacy Fluentd collector. However, Fluentd is deprecated and slated for removal in future releases. Red Hat